

Ex. No.: 11c)
Date:15/4/25

Name:Rithupriya S
Roll no:231901043

Optimal

Aim:

To write a c program to implement Optimal page replacement algorithm.

ALGORITHM:

1. Start the process
2. Declare the size
3. Get the number of pages to be inserted
4. Get the value
5. Declare counter and stack
6. Select the least frequently used page by counter value
7. Stack them according the selection.
8. Display the values
9. Stop the process

PROGRAM:

```
#include <stdio.h>
```

```
int search(int key, int frame[], int f) {    for (int i = 0; i < f; i++) {        if  
(frame[i] == key)            return 1;  
    }    return 0;  
}
```

```
int predict(int pages[], int frame[], int n, int index, int f) {    int res = -1, farthest = index;
```

```
    for (int i = 0; i < f; i++) {        int j;  
        for (j = index; j < n; j++) {            if (frame[i] == pages[j]) {                if  
(j > farthest) {                    farthest = j;                    res = i;                }  
            }  
        }  
    }
```

```
    // If page not found in future, return that index  
    if (j == n)        return i;  
}
```

```

    return (res == -1) ? 0 : res;
}

int main() {
    int n, f;
    printf("Enter number of frames: ");    scanf("%d", &f);

    printf("Enter number of pages: ");    scanf("%d", &n);

    int pages[n];    printf("Enter reference string: ");    for (int i = 0; i < n; i++)
        scanf("%d", &pages[i]);

    int frame[f];
    int count = 0, index = 0;

    for (int i = 0; i < f; i++)
        frame[i] = -1;

    for (int i = 0; i < n; i++) {        if (search(pages[i], frame, f)) {
        // No page fault        } else {        if (index < f) {
frame[index++] = pages[i];
        } else {
            int pos = predict(pages, frame, n, i + 1, f);        frame[pos] = pages[i];
        }        count++;
        }

    for (int j = 0; j < f; j++) {        if (frame[j] != -1)        printf("%d
", frame[j]);
        else        printf("-1 ");
    }    printf("\n");
}

printf("\nTotal Page Faults = %d\n", count);
return 0;
}

```

Output:

```

Enter number of frames: 3
Enter number of pages: 12
Enter reference string: 7 0 1 2 0 3 0 4 2 3 0 3

```

```

7 -1 -1
7 0 -1
7 0 1
2 0 1
2 0 1 2 0 3

```

2 0 3

4 0 3

4 0 2 4 3 2

0 3 2

0 3 2

Total Page Faults = 9

Result:

Thus the algorithm is executed successfully.